

NCBA Country Info Service — Kubernetes Troubleshooting Guide

Step 10 — Troubleshooting the application on Kubernetes

NCBA Integration Microservices

Kubernetes Troubleshooting Guide

This guide helps diagnose and resolve common runtime problems for the **Country Info microservice** running in the `ncba-countryinfo` namespace.

All commands assume the namespace flag `-n ncba-countryinfo`. Set it once with `kubectl config set-context --current --namespace=ncba-countryinfo` to omit it.

0. First-line triage

Start every investigation with a broad snapshot, then drill in:

```
kubectl -n ncba-countryinfo get pods,svc,deploy,statefulset,hpa,ingress
kubectl -n ncba-countryinfo get events --sort-by=.lastTimestamp | tail -30
kubectl -n ncba-countryinfo describe pod <pod> # 'Events:' section at the bottom
kubectl -n ncba-countryinfo logs <pod> --tail=200
kubectl -n ncba-countryinfo logs <pod> --previous # logs from a crashed container
```

The application emits **structured JSON logs** (prod profile), so you can filter:

```
kubectl -n ncba-countryinfo logs deploy/countryinfo-app | grep '"level":"ERROR"'
```

1. Pod in `CrashLoopBackOff`

The container starts, exits, and Kubernetes keeps restarting it.

Diagnose

```
kubectl -n ncba-countryinfo describe pod <pod>          # look at Events + Last State
      (exit code)
kubectl -n ncba-countryinfo logs <pod> --previous      # the actual stack trace / startup
      error
```

Common causes & fixes

Symptom in logs	Cause	Fix
Communications link failure / Unknown database	DB not reachable at startup	Ensure MySQL pod is Ready; check DB_HOST=mysql, credentials in Secret
Access denied for user	Wrong DB credentials	Reconcile secret.yaml with what MySQL was initialised with
OOMKilled (in describe, Last State)	Heap exceeds memory limit	Raise memory limit, or lower JAVA_OPTS MaxRAMPercentage
Port 8080 already in use	Misconfiguration	Usually transient on restart; check no custom SERVER_PORT clash

The liveness probe has a `startupProbe` guard (up to ~5 min) so a slow first boot is not mistaken for a crash. If a pod still crash-loops, it is a genuine startup failure — read `--previous` logs.

2. Pod stuck in Pending

The pod is scheduled but cannot start.

Diagnose

```
kubectl -n ncba-countryinfo describe pod <pod>          # reason is in Events
kubectl -n ncba-countryinfo get pvc                    # for the MySQL StatefulSet
kubectl get nodes -o wide
kubectl describe node <node> | grep -A5 Allocated
```

Common causes & fixes

Event message	Cause	Fix
Insufficient cpu / Insufficient memory	No node has the requested resources	Lower <code>resources.requests</code> , or add/scale nodes
pod has unbound immediate PersistentVolumeClaims	PVC cannot bind	Ensure a default <code>StorageClass</code> exists (<code>kubectl get storageclass</code>); on bare clusters install one (e.g. local-path)
0/N nodes are available: node(s) had taint	Taints/affinity	Add tolerations or deploy to a schedulable node

3. Service unreachable / no response

The app pods run, but requests to the Service time out or return nothing.

Diagnose

```
kubectl -n ncba-countryinfo get endpoints countryinfo-app # MUST list pod IPs
kubectl -n ncba-countryinfo get pods --show-labels
kubectl -n ncba-countryinfo describe svc countryinfo-app
```

Checklist

- **Empty ENDPOINTS** → the Service selector (`app: countryinfo-app`) does not match pod labels, or no pod is *Ready* (only Ready pods are added to endpoints — check readiness probes).
- **Port mismatch** → Service `targetPort` must be `8080` (the container port); the Service `port` is `80`.
- **Test from inside the cluster** to isolate ingress vs. service issues:

```
kubectl -n ncba-countryinfo run tmp --rm -it --image=busybox --restart=Never -- \
  wget -q0- http://countryinfo-app/actuator/health
```

4. Database connection refused

The app cannot reach MySQL.

Diagnose

```
kubectl -n ncba-countryinfo get pods -l app=mysql
kubectl -n ncba-countryinfo logs statefulset/mysql --tail=100
kubectl -n ncba-countryinfo get svc mysql # headless service must exist
kubectl -n ncba-countryinfo get configmap countryinfo-config -o jsonpath='{.data.DB_HOST}'; echo
kubectl -n ncba-countryinfo get secret countryinfo-secret -o jsonpath='{.data.DB_PASSWORD}' | base64 -d; echo
```

Checklist

- MySQL pod **Ready**? If 0/1, read its logs — first boot initialises the DB and can take 30–60s; the app's startup probe tolerates this.
- `DB_HOST` in the ConfigMap must equal the MySQL Service name (`mysql`).
- Secret values (`DB_USERNAME`, `DB_PASSWORD`) must match the values MySQL was created with. If you changed them after first boot, the existing PVC still has the old credentials — recreate the DB (`delete pvc`) or fix the user.
- Connectivity test:

```
kubectl -n ncba-countryinfo exec -it statefulset/mysql -- \
mysql -u appuser -p"$DB_PASSWORD" -e "SELECT 1;"
```

5. High CPU / memory (and autoscaling)

Diagnose

```
kubectl -n ncba-countryinfo top pods # needs metrics-server
kubectl -n ncba-countryinfo get hpa countryinfo-app
kubectl -n ncba-countryinfo describe hpa countryinfo-app
```

Checklist

- **HPA shows <unknown>/70%** → metrics-server is not installed or not ready. Install it (`kubectl apply -f https://github.com/kubernetes-sigs/metrics-server/...`) and confirm `kubectl top pods` works.
- **Pods pinned at the CPU limit (500m)** under load → the HPA should add replicas up to 10. If it is not scaling, check the HPA conditions in `describe hpa` (`AbleToScale`, `ScalingActive`).
- **Frequent OOM restarts** → raise the memory limit, or reduce JVM heap via `JAVA_OPTS` (`-XX:MaxRAMPercentage`).

- **External SOAP latency** inflating request threads → the SOAP client has 5s/10s timeouts + circuit breaker; inspect breaker state at `/actuator/health` and `resilience4j_*` metrics at `/actuator/prometheus`.

6. Rolling update stuck

A new revision will not finish rolling out.

Diagnose

```
kubectl -n ncba-countryinfo rollout status deployment/countryinfo-app
kubectl -n ncba-countryinfo get pods                # new ReplicaSet pods not becoming Ready?
kubectl -n ncba-countryinfo describe pod <new-pod>  # readiness probe failing? image pull error?
```

Common causes & fixes

Cause	How to confirm	Fix
New image fails readiness	<code>describe pod</code> → readiness probe errors	Fix the image/config; rollout halts safely (<code>maxUnavailable: 0</code> keeps old pods serving)
<code>ImagePullBackOff</code>	<code>describe pod</code> Events	Wrong image name/tag or registry auth; fix <code>image: /</code> <code>imagePullSecret</code>
Bad config in new revision	new pods crash-loop	Roll back, then fix

Roll back to the last working revision:

```
kubectl -n ncba-countryinfo rollout undo deployment/countryinfo-app
kubectl -n ncba-countryinfo rollout history deployment/countryinfo-app
```

7. Ingress not routing

```
kubectl -n ncba-countryinfo describe ingress countryinfo-ingress
kubectl get pods -n ingress-nginx                # controller installed & running?
```

- Confirm an ingress controller is installed and the `ingressClassName: nginx` matches it.

- Ensure the request `Host` header matches `countryinfo.local` (or your host), and that the host resolves to the ingress address (`/etc/hosts` or DNS).
 - Test the Service directly (port-forward) to confirm the problem is the ingress layer, not the app.
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8. Useful one-liners

Tail only error logs across all app pods

```
kubectl -n ncba-countryinfo logs -l app=countryinfo-app --tail=-1 -f | grep ERROR
```

Exec a shell into a running app pod

```
kubectl -n ncba-countryinfo exec -it deploy/countryinfo-app -- sh
```

Watch pods during an incident

```
kubectl -n ncba-countryinfo get pods -w
```

Recent events, newest last

```
kubectl -n ncba-countryinfo get events --sort-by=.lastTimestamp
```

Restart the app cleanly (e.g. after a config change)

```
kubectl -n ncba-countryinfo rollout restart deployment/countryinfo-app
```

For deployment steps, see the companion **Kubernetes Deployment Guide**.